

Closest point method with an angle rotation glue: convergence of the L_∞ error on two creased surfaces The flipped cap on an ellipsoid, and the half-twisted rectangular tube

Results from `cp_matrices/examples_2026/3D_surface`

October 5, 2026

Abstract

This report explains two three-dimensional tests of the closest point method with an angle rotation glue at a surface junction. Both tests solve $u - \Delta_S u = f$ on a surface made of two branches that meet along a closed curve. The first test cuts an ellipsoid along a circle and folds the cap inside the bowl, which makes a crease along the whole circle. The second test splits a half-twisted rectangular tube into two strips along both of its corner curves. The report describes how the grid and the two bands are built, how the glue rotation is found and applied, and how the discrete operator is formed. It then gives the measured L_∞ convergence rates. **Headline result:** neither surface recovers the second-order rate of a smooth closest point solve. Every fitted rate lies between 1.2 and 1.8. On the flipped ellipsoid the *size* of the error at a fixed grid spacing grows with the curvature mismatch $\delta\kappa$ at the crease, although the fitted rate does not order by $\delta\kappa$ over the number of levels run here. Without the glue, the same solve does not converge at all. On both surfaces the rotation may be estimated from closest point differences alone, at no cost in the solution, and the junction transmission converges faster than the solution does.

Contents

1	Summary of the results	2
2	What the two tests are	3
2.1	Test 1: the flipped cap on an ellipsoid	4
2.2	Test 2: the half-twisted rectangular tube	4
3	How we build the grid	5
3.1	Bandwidth	5
3.2	The box and the grid offsets	5
3.3	Finding the nodes near the surface	5
3.4	From candidates to bands	6
3.5	The grid spacings	6
4	How we find and apply the rotation	7
4.1	Which rows are rotated	7
4.2	The rotation axis	7
4.3	The angle	7
4.4	The construction: Rodrigues, not <code>atan2</code>	8
4.5	The estimated rotations	8
4.6	Applying the rotation	9
4.7	Measuring the rotation error	9
5	How we form the operator	9
5.1	Test 1: the vGMM operator	9
5.2	Test 2: the stabilized (diagonally split) ICPM operator	10
5.3	What the scripts refuse to do	11

6	How we measure the error	11
7	Results: the flipped cap on an ellipsoid	12
7.1	The geometry of one cut	12
7.2	The glue is what makes the solve converge	12
7.3	Why a level-to-level rate is meaningless here	13
7.4	The cross-branch measures	13
7.5	The estimated rotations	14
7.6	All four cuts, against the curvature mismatch	14
8	Results: the half-twisted rectangular tube	15
8.1	The bands and the routed rows	15
8.2	Convergence of the L_∞ error	18
8.3	The estimated rotation costs nothing here	18
8.4	The traces agree at nearly second order	18
9	What the two tests say together	19
9.1	The bound the two tests are read against	19
9.2	The two surfaces are different tests of the same thing	20
9.3	Why the ellipsoid scatters and the tube does not	20
9.4	What would settle the open question	21
9.5	A note on the construction, separate from the results	21
10	How to reproduce	21
10.1	Two changes were needed to produce these runs	21

1 Summary of the results

The question this report answers is: *what convergence rate does the L_∞ error reach, when the closest point method is combined with an angle rotation glue at a surface junction?*

The answer, on both surfaces, is **between first and second order, and closer to first**. Table 1 collects it.

Table 1: The headline numbers. Every rate is a least-squares slope of $\log \|e\|_\infty$ against $\log h$ over all the levels of that run.

test	variant	levels	fitted rate	error at the finest h
<i>Flipped cap on an ellipsoid, $h = 0.065 \rightarrow 0.0241$</i>				
cut $z = 0.390$, $\delta\kappa = 3.07$	exact R	6	1.26	6.44e-03
cut $z = 0.455$, $\delta\kappa = 2.68$	exact R	6	1.80	5.62e-03
cut $z = 0.520$, $\delta\kappa = 2.35$	exact R	6	1.47	3.75e-03
cut $z = 0.585$, $\delta\kappa = 2.07$	exact R	6	1.23	3.24e-03
cut $z = 0.390$, $\delta\kappa = 3.07$	<i>no glue</i>	6	0.10	2.84e-01
<i>Half-twisted rectangular tube, $h = 1/20 \rightarrow 1/44$</i>				
both corner curves	exact rotation	5	1.43	4.42e-02
both corner curves	$d_{\kappa 2}$ rotation	5	1.43	4.42e-02

The five findings

1. The glue is what makes the solve converge. On the flipped ellipsoid, the same closest point method applied to the creased surface *without* the glue does not converge at all. Its fitted slope is 0.10 and its error stays near 0.3 at every grid spacing. The glued solve is 44 times more accurate at the same grid spacing, and it converges. This is the largest effect in the whole study.

2. Neither surface reaches second order. Every fitted rate lies between 1.2 and 1.8. This is what the surface bound predicts when the junction defect does not vanish: the term $h(D + J_2)$ is first order and dominates the h^2 interior term. On the tube the level-to-level rates fall steadily through the refinement ($2.23 \rightarrow 1.59 \rightarrow 0.86 \rightarrow 1.05$), so the fitted 1.43 is an average over a range in which the rate is still moving towards first order, rather than an order the scheme holds.

3. The glue rotation does not have to be accurate. The rotations built only from closest point differences give the same solution as the exact analytic rotation, everywhere in this study. On the tube the two agree to 0.16 per cent at worst, although the estimated rotation is wrong by 1.40 rad (80°) on its worst row at the coarsest level. On the ellipsoid the three variants agree to within 1 per cent at the finest level of every cut. The glue has to be approximately right; it does not have to be exact.

4. The transmission is not what limits the solve. On both surfaces the two branches agree with *each other* across the junction at a higher rate than either agrees with the exact solution: 1.93 and 1.71 for the tube's two corner curves against 1.43 for its surface error, and 1.97 for the ellipsoid's cross-branch gap against 1.26 for its surface error. The glue transmits consistently. The limit is the geometry of the junction itself.

5. On the ellipsoid, the error is not a smooth function of h . Neighbouring grid spacings differ by a factor of three, and level-to-level rates run from -6.3 to $+6.8$, while the linear solver converges to 10^{-10} at every level. The cause is a signed cancellation of the junction residual around the cut circle, whose outcome depends on where that circle falls between grid planes. Only the fitted slope is a rate. The tube does not show this, because its junction curves are not plane circles and its glue angle varies along them.

What to be careful about

- **The fits are short.** The ellipsoid script's default is ten grid spacings, chosen to average over the scatter in finding 5. The runs here use six, because the finer levels of the default list cost more than the time available. The ellipsoid rates should be read as "first order, not second", not as three-digit numbers. See Section 7.3.
- **The $\delta\kappa = 0$ control was not run.** The script can place the cut without flipping the cap, which makes the curvature mismatch vanish identically and should recover second order. That run would separate "the crease costs an order" from "the glue costs an order". It is the obvious next step.
- **The error level orders by $\delta\kappa$; the fitted rate does not.** The four cut heights give rates 1.26, 1.80, 1.47 and 1.23, which are not monotone in $\delta\kappa$. The errors at a fixed grid spacing are monotone. See Section 7.6.
- **A bug blocks the script as committed.** `error_surface_figure` uses `thetaex`, which is never passed to it, so the ellipsoid script aborts after its first cut whenever `z_3d` is not empty. The runs here use a copy with that argument added. See Section 10.

2 What the two tests are

Both tests solve the same screened problem on a surface in $\mathbb{R}^3$,

$$u - \Delta_S u = f, \tag{1}$$

with a manufactured solution. Both surfaces are made of two pieces. The pieces meet along a closed curve. The surface is not smooth along that curve.

A single closest point extension is not available at such a curve. The medial axis of the surface comes out of the curve. A stencil that crosses the medial axis couples the two pieces directly. The closest point map is not single valued there.

Each test therefore uses one band per piece. We call a piece a *branch*. Each band is complete except at the rows whose own closest point falls on the shared curve. Those rows have no footpoint of their own to read. The *angle rotation glue* supplies them. It turns the grid node rigidly about the shared curve, then projects the turned node onto the other branch.

2.1 Test 1: the flipped cap on an ellipsoid

The surface is the ellipsoid of revolution about the z axis,

$$\gamma(\varphi, \vartheta) = (a \sin \varphi \cos \vartheta, a \sin \varphi \sin \vartheta, c \cos \varphi), \quad \varphi \in [0, \pi], \quad (2)$$

with $a = 1$ and $c = 0.65$. A plane $z = z_{\text{cut}}$ cuts it. Branch 1 is the bowl below the cut. Branch 2 is the cap above the cut.

The cap is then reflected down across the cut plane, $(x, y, z) \mapsto (x, y, 2z_{\text{cut}} - z)$. The cap becomes a dimple inside the bowl. The two pieces now meet along the whole cut circle in a crease. This is the **flipped** case, and it is the default.

A reflection is an isometry. The flipped surface therefore carries the same metric as the ellipsoid in the same (φ, ϑ) . The manufactured solution and its Laplace–Beltrami operator are the same functions of (φ, ϑ) on both branches. The errors of the two branches are directly comparable, and the jump $[f]$ is zero.

The manufactured solution is

$$u = \cos \varphi + \frac{1}{2} \cos 2\varphi + B \sin^2 \varphi \cos 2\vartheta, \quad B = 0.4, \quad (3)$$

which is a polynomial in z plus $B(x^2 - y^2)/a^2$. It is therefore smooth at both poles, where the (φ, ϑ) chart is not. $\Delta_S u$ is analytic.

The script runs four cut heights, written through $\sin t_c = (z_{\text{cut}} - z_{\text{cen}})/c$:

$$s_0 = \sin t_c \in \{0.6, 0.7, 0.8, 0.9\}. \quad (4)$$

The crease angle and the curvature mismatch grow together as the cut slides from the pole towards the equator. The list is therefore a list of curvature mismatches.

2.2 Test 2: the half-twisted rectangular tube

The surface is a rectangular tube whose cross-section turns by half a turn around the centreline circle,

$$F(\theta, p) = (R + u)(\cos \theta, \sin \theta, 0) + v(0, 0, 1), \quad \begin{aligned} u &= p_1 \cos \frac{\theta}{2} - p_2 \sin \frac{\theta}{2}, \\ v &= p_1 \sin \frac{\theta}{2} + p_2 \cos \frac{\theta}{2}, \end{aligned} \quad (5)$$

with $R = 1$, $a = 0.45$ and $b = 0.20$. The angle θ is 4π periodic. Branch A (wide) is $p = (s, b)$ with $s \in [-a, a]$. Branch B (narrow) is $p = (a, s)$ with $s \in [-b, b]$. Each strip covers its face pair once over $\theta \in [0, 4\pi)$.

The two strips meet along *two* closed curves:

$$\text{upper: } A \text{ at } (\theta, +a) = B \text{ at } (\theta, +b), \quad (6)$$

$$\text{lower: } A \text{ at } (\theta, -a) = B \text{ at } (\theta + 2\pi, -b) \pmod{4\pi}. \quad (7)$$

The shift by 2π on the lower curve is the half twist. The complete surface is a closed orientable torus. Figure 4 shows it.

The metric of a strip is $E = |F_s|^2 = 1$, $H = F_s \cdot F_\theta$ (equal to $-b/2$ on A and $+a/2$ on B), and $G = |F_\theta|^2$. The two faces are not exactly perpendicular, because of the twist. The glue angle

therefore varies along each curve. This is the main difference from the ellipsoid, where one angle serves the whole cut circle.

The manufactured solution is, on each strip, the cubic Hermite interpolant in s between edge traces and edge slopes. The traces and slopes vary smoothly in θ . Both strips use the same trace on each curve. The slopes are chosen so that the two outgoing conormal fluxes sum to zero on both curves. The traces therefore agree and the fluxes balance, which is what the transmission conditions ask for. The forcing $f = u - \Delta_S u$ is analytic to rounding.

3 How we build the grid

Both tests use *one* Cartesian grid and *two* bands on it. The shared grid is what makes the cross-branch block of the operator assemblable. With one grid per branch there is no common column index space, and an extra grid-to-grid transfer would pollute the glue error we want to measure.

3.1 Bandwidth

Both tests take the bandwidth from the Ruuth–Merriman formula [2],

$$\text{bw} = 1.0002 \sqrt{(d-1) \left(\frac{p+1}{2} \right)^2 + \left(\frac{q}{2} + \frac{p+1}{2} \right)^2}, \quad (8)$$

with $d = 3$, interpolation degree $p = 3$ and Laplacian order $q = 2$. This gives $\text{bw} \approx 4.12$. The factor 1.0002 is a safety factor. The two tangential directions of a surface each contribute one interpolation half-stencil. The normal direction contributes one interpolation half-stencil plus one Laplacian half-stencil. That is what the composition EL reads.

3.2 The box and the grid offsets

The box is the bounding box of the two pieces *as they are embedded*, plus a pad of a few cells. For the flipped ellipsoid the box is much shorter in z than the ellipsoid’s own box, because the cap is folded inside the bowl. For the tube the box comes from the centreline radius R plus the cross-section radius $\rho = \sqrt{a^2 + b^2}$.

Each axis is then shifted by an odd fraction of the grid spacing:

Test	shift of (x, y, z) in units of h
flipped ellipsoid	$(-0.317, -0.211, -0.133)$
half-twisted tube	$(+0.317, +0.211, +0.173)$

The shifts matter. A grid that is symmetric about the cut plane gives the two branches mirror-image tangent estimates. The errors of those estimates then cancel in the glue rotation, and the estimator looks better than it is. The shifts also keep the junction curve off the grid nodes.

3.3 Finding the nodes near the surface

A full `meshgrid` over the box cannot be built at the finest grid spacing, and the banding would discard all but a shell of it. Each test therefore prefilters first.

Ellipsoid. The script walks each piece in the polar angle φ , and at each φ around the parallel with a count set by that parallel’s own circumference. The poles are therefore not oversampled. Samples are no more than about $h/2$ apart. Each sample is snapped to the nearest grid node. The seed set is then grown by $\text{rad} = \lceil \text{bw} \rceil + 2$ cells, one axis at a time, with unique linear indices taken after each axis. The working set stays the size of the band instead of $(2\text{rad} + 1)^3$ times it.

Tube. Every strip point lies within ρ of the centreline circle. A node within bw h of either strip is therefore within $\rho + \text{bw } h$ of that circle. The prefilter keeps the nodes with

$$(\sqrt{x^2 + y^2} - R)^2 + z^2 \leq (\rho + \text{bw } h)^2. \quad (9)$$

This is the *conservative initial band*. Everything the operator needs must lie inside it, or the run stops with an error.

3.4 From candidates to bands

Each branch then runs its *own* closest point function over the candidate nodes. The ellipsoid uses `cpEllipseArc` in the meridian half plane, spun back out along the azimuth of the query; a reflected piece is handled by reflecting the query, solving, and reflecting the answer. The tube uses `cpHalfTwistedRectTube`, which clamps s exactly for fixed θ and then refines θ by safeguarded Newton iteration from 256 periodic samples.

A node is kept when $|\text{dist}| \leq \text{bw } h$. Near the junction curve a node lies in *both* bands. It then carries *two* independent unknowns, one per branch field. The bands overlap in space; the unknowns never overlap.

The two tests then differ in how the band is closed.

Ellipsoid: one band per branch. The vGMM operator applies L to the band values and extends the result. E and L are therefore both square on the same node set, and no restriction operator is needed. The script marks the rows of L whose whole stencil landed in the band (`1full`). A row at the outer edge loses part of its stencil. That is harmless while nothing reads the row, and the script checks that: in EL a row of L is read only when E has a nonzero in that column.

Tube: inner band, outer band, and a closure sweep. The inner band of a branch must hold every interpolation stencil used on it. Three sets of stencils must fit:

1. the stencils of the closest points of its own initial band;
2. the stencils of every surface error sample and every corner-curve trace sample;
3. the stencils of the target closest points of the rows routed *into* it, by *either* rotation.

The outer band is the inner band plus its 7-point Laplacian neighbours. Enlarging an inner band enlarges the outer band. A larger outer band can create new routed rows, and those rows need new support. Banding and routing are therefore iterated until nothing new appears. The sweep count is reported. Both rotation methods share one set of bands and one set of routed rows, so only the angle differs between them.

3.5 The grid spacings

The flipped ellipsoid uses $h_k = 0.065 \times 0.82^k$. A band node has a unique closest point on its branch only while $\text{bw } h$ stays below the smallest radius of curvature, which on an oblate ellipsoid is c^2/a at the equator. The script keeps only the h that satisfy $\text{bw } h < 0.65 \min(c^2/a, a^2/c)$.

Many grid spacings are needed, and the reason is specific to this surface. The surface error of the flipped solve is *not* a smooth function of h . Its size is set by the net signed junction residual summed around the cut circle. That sum is a near cancellation: each branch contributes a systematic total of one sign, the other branch the opposite sign. What survives depends on where the cut circle falls between grid planes. Neighbouring h can differ by a factor of several, and a level-to-level rate is meaningless. Only the least-squares slope through many levels is a rate. In the plane the same cancellation exists, but the junction is two points instead of a circle, the sums run over $O(1)$ rows instead of $O(1/h)$ rows, and the noise never surfaces.

The tube uses an explicit list of h , and rates use the actual ratios.

4 How we find and apply the rotation

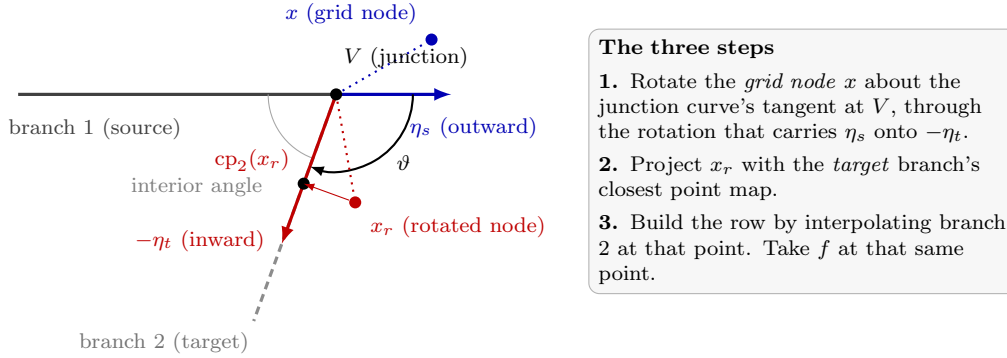


Figure 1: The angle rotation glue, drawn in the plane normal to the junction curve. Here η_t is the target branch's *outward* conormal, so branch 2 itself lies along $-\eta_t$. The rotation carries the source branch's outward direction η_s onto $-\eta_t$. Leaving one branch means entering the other, so outward-onto-inward is the pairing a rigid continuation needs. The object that is rotated is the grid node x , not its closest point. This carries both the overhang past the junction and the normal offset across intact. On the ellipsoid this plane is a meridian half plane, and one angle serves the whole cut circle. On the tube the plane is the normal plane of a corner curve, and the angle varies along the curve, because the twist stops the two faces from being perpendicular.

4.1 Which rows are rotated

A row is rotated when its own closest point lies on the junction curve.

On the ellipsoid, `cpEllipseArc` returns the clamped end of the meridian arc exactly. The test is therefore done on the meridian coordinates (ρ, z) of the closest point, with the reflection undone first, to a tolerance of $10^4 \varepsilon$. The poles are the other ends of the two meridian arcs. They are not boundaries, because the surface of revolution closes up there, and they are correctly not matched.

On the tube, the closest point function returns a flag $\text{bdy} = \pm 1$ when the closest point lies on the edge $s = \mp W$. Every outer row with $\text{bdy} \neq 0$ is routed.

4.2 The rotation axis

The axis is the tangent to the junction curve at the row's own closest point.

On the ellipsoid that tangent is the azimuthal direction e_ϑ . Two facts make this the honest lift of the planar construction. First, the closest point V of the node lies on a *circle*, so the offset of the node from V has no e_ϑ component; the node lies in V 's own meridian half plane, and a rotation about e_ϑ keeps it there. Second, the surface is one of revolution, so the closest point of the rotated node on the other branch lies in the same half plane. The whole construction is the planar one, applied meridian by meridian, and the angle is the planar angle in the (ρ, z) coordinates of that half plane.

On the tube the axis is $\tau = F_\theta / |F_\theta|$ at the edge closest point. Both strips share it. Both methods use the analytic τ .

4.3 The angle

The glue carries the *outward* direction of the source branch onto the *inward* direction of the target branch. Leaving the source branch at the junction means entering the target branch, so outward-onto-inward is the pairing a rigid continuation needs. On a smooth join the two directions are opposite and the rotation is the identity.

On the ellipsoid the two directions are the outward unit tangents of the two meridian arcs at the cut. Reflecting branch 2 negates the z component of its tangent, and that is what opens the crease. At $s_0 = 0.6$, for example, the exact glue angle is 1.4282 rad (81.8°).

On the tube the two directions are the outward conormals,

$$\eta_s = \text{sign}(s) \text{unit}(F_s - (F_s \cdot \tau)\tau), \quad (10)$$

on the source strip, and η_t likewise on the target strip at the glued parameters. The glue carries η_s onto $-\eta_t$. Because of the twist this angle varies along the curve, so it is computed per row.

4.4 The construction: Rodrigues, not atan2

Nothing in this code stores a glue angle. The rotation is stored as a $(\cos, \sin)$ pair, and the pair comes straight out of the two directions. Rodrigues' formula about a unit axis k is

$$R = I + \sin \vartheta K + (1 - \cos \vartheta) K^2, \quad K = \text{skew}(k). \quad (11)$$

In the plane the axis is e_z , $K^2 = -I$, and the formula collapses to $R = \cos \vartheta I + \sin \vartheta K = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ with

$$c = u \cdot w, \quad s = (u \times w) \cdot e_z = u_1 w_2 - u_2 w_1, \quad (12)$$

followed by one division by $\text{hypot}(c, s)$. In three dimensions the same two quantities are $c = u \cdot w$ and $s = \tau \cdot (u \times w)$. No angle is formed and no transcendental function is evaluated.

The angle-first alternative reaches the same matrix through six library transcendental calls, and $\cos$ and $\sin$ immediately undo what `atan2` just did. Four consequences follow, and the planar companion experiment measures all of them:

- **Accuracy.** Against a rotation known to full precision, $\|R - R_{\text{exact}}\|$ comes out two to seven times larger for the angle-first path at every angle sampled over a full turn. The worst case over that sweep is 1.2×10^{-15} against 2.8×10^{-16} . Both are a few units in the last place, so this is a reason to prefer the construction, not a defect being repaired.
- **No branch cut.** A difference of two `atan2` values lands in $(-2\pi, 2\pi]$ and must be wrapped back. The stored number is then discontinuous at $\pm\pi$, although the rotation it stands for is not. A $(\cos, \sin)$ pair has no such seam.
- **Exact inverses.** Swapping the two directions gives the other branch's glue at the same junction point. Under Rodrigues the cosine comes out bit for bit the same and the sine bit for bit negated, so the two branches' rotations are *exact* transposes. The angle-first path satisfies that only to within rounding.
- **Reproducibility.** IEEE 754 pins down multiply, add, divide and `hypot`. It says nothing about `atan2`, `sin` or `cos`. Dot products, cross products and `hypot` cannot differ between platforms; the angle-first matrix can.

4.5 The estimated rotations

The exact rotation uses analytic directions. It is the reference. Each test also runs rotations built only from closest point differences, because a real application has no analytic frame.

Write cp for the closest point of the node x on its own branch, and reflect the node through it:

$$\overline{\text{cp}} = \text{cp}(2 \text{cp} - x), \quad \overline{\overline{\text{cp}}} = \text{cp}(3 \text{cp} - 2x). \quad (13)$$

The two estimators are

$$d_k = \text{cp} - \overline{\text{cp}} \quad (\text{first order}), \quad (14)$$

$$d_{k2} = 1.5 \text{cp} - 2 \overline{\text{cp}} + 0.5 \overline{\overline{\text{cp}}} \quad (\text{second order}). \quad (15)$$

A plain difference of closest points returns nothing at these rows, because the closest point *is* the junction point. That is why the node is reflected first.

Ellipsoid. Each difference is resolved in its own node's meridian frame (e_ρ, e_z) *before* averaging. By rotational symmetry that is the frame the glue angle is measured in. Averaging the raw vectors instead would cancel them around the circle. The e_ϑ component is discarded; it is zero for the exact tangent and only noise here. The script runs three variants per cut: d_k , d_{k2} , and the exact rotation.

Tube. Both conormals are estimated by d_{k2} , projected off the exact τ and normalized. The source estimate reflects the original routed grid node, $v = c - x$. A row is re-sampled with a deterministic outward probe $x' = c + h\eta_s$ when any of these holds: $|v|$ is tiny; the projected difference is short relative to $|v|$ ($\alpha = |d_\perp|/|v| < 0.25$); a reflected closest point is itself on an edge; the sample jumps to the opposite sheet; or the estimate points inward. The target estimate always uses the outward probe $c_t + h\eta_t$. An unusable probe raises an error; there is no fallback to the exact conormal.

This makes the tube's d_{k2} *not* a fully geometry-free estimator, and the script says so. The exact τ supplies the axis and the projection plane. The exact conormals place the probes and check the outward orientation. Both directions that set the angle are still closest point differences.

4.6 Applying the rotation

We rotate the *grid node*, never its closest point. That is what makes the map a rigid continuation: both the tangential overhang past the junction and the normal offset are carried across intact. Rotating the closest point instead would only reparametrize the junction curve, and it would lose the normal coordinate.

On the ellipsoid, the offset $x - V$ is resolved in the node's own meridian frame into $(d_\rho, d_\vartheta, d_z)$. The planar rotation is applied to (d_ρ, d_z) . The component d_ϑ is zero up to rounding, and it is carried through anyway, so the formula does not quietly depend on it vanishing. The result is mapped back to Cartesian coordinates.

On the tube, Rodrigues' formula is applied directly,

$$x_r = c + (x - c) \cos \vartheta + (\tau \times (x - c)) \sin \vartheta + \tau (\tau \cdot (x - c)) (1 - \cos \vartheta), \quad (16)$$

again from the $(\cos, \sin)$ pair and not from an angle.

The rotated node is then projected with the *target* branch's closest point function. The interpolation stencil is built there.

4.7 Measuring the rotation error

The error of an estimated rotation is reported through the *relative* rotation $R_{\text{exact}}^\top R$, whose angle lands in $(-\pi, \pi]$ by construction:

$$c_{\text{rel}} = c_{\text{ref}} + s_{\text{ref}}, \quad s_{\text{rel}} = s_{\text{ref}} - c_{\text{ref}}. \quad (17)$$

There is nothing to re-wrap. The obvious alternative, a wrapped difference of two angles, loses digits exactly where the comparison matters, which is when both angles sit near $\pm\pi$.

5 How we form the operator

5.1 Test 1: the vGMM operator

The ellipsoid test uses the embedded method-of-lines operator of von Glehn, März and Macdonald [1],

$$M = EL - \gamma(I - E), \quad \gamma = \frac{2d}{h^2}, \quad (18)$$

and solves $(I - M)u = f$. Here E is the degree-3 interpolation at the closest points, square on the band, and L is the second-order 7-point Laplacian on the same band.

The glue needs no further handling, and that is the attraction of this operator. L is block diagonal, so

$$(EL)_{k,o} = E_{k,o} L_o \quad (19)$$

extends the *other* branch's Lu across the cut. The stabilization row $u_i - (Eu)_i$ at a rotated node of branch k is exactly the statement that branch k 's value there agrees with branch o 's interpolant at the rotated point. In assembly, a routed row's entry in the diagonal block is zeroed, and its weights go into the off-diagonal block.

M is never assembled. In three dimensions EL has some hundreds of entries per row; E and L separately have 64 and 7. The operator is applied as

$$(I - M)v = (1 + \gamma)v - E(Lv + \gamma v), \quad (20)$$

under restarted GMRES with a Jacobi preconditioner built from $\text{diag}(EL)_i = \sum_j E_{ij} L_{ji}$. The tolerance is 10^{-10} , the restart is 40, and the cap is 400 restart cycles.

The forcing at a junction row. A rotated row's unknown does not stand for u at that row's own closest point. That point is on the cut circle. It stands for u at the rotated target on the other branch, which is up to $bw h$ of arclength away. Its forcing is therefore f at the *rotated target*.

This is the one place where the three-dimensional script deliberately differs from its planar companion, and the reason is a counting argument. Using f at the row's own clamped closest point leaves a residual $f(\text{target}) - f(V) = O(h)$ on every one of those rows. With $O(1)$ such rows in the plane that is invisible. With $O(1/h)$ of them around a circle it is not.

That choice needs a caveat, because it is the convention the surface analysis singles out as the dangerous one. Evaluating the forcing on the opposite side without correcting a trace jump can leave the forcing residual at $O(1)$ instead of $O(h)$, and that would promote its contribution from $O(h^2)$ to $O(h)$. It is safe here only because $[f] = 0$: u and f are the same functions of (φ, ϑ) on both branches, so the two one-sided traces agree and there is no jump to correct. A forcing written per branch rather than per (φ, ϑ) would break that, and would then need f at the source-side edge point or an explicit trace correction.

The baseline. The same vGMM operator is also run on *one* band over the union of the two pieces. On the unflipped surface that is the plain ellipsoid, and it measures the uncut rate. On the flipped surface it is the closest point method applied to the crease *as if the crease were not there*. That is the other number worth having.

5.2 Test 2: the stabilized (diagonally split) ICPM operator

The tube test uses the diagonally split operator. Let L be the 7-point Laplacian from the inner band rows to the outer band columns, and R the restriction from outer to inner. Split L by its own diagonal,

$$L_{\text{diag}} = \sum_j (R \odot L)_{ij}, \quad L_{\text{off}} = L - R \odot L, \quad (21)$$

and form

$$M = \text{diag}(L_{\text{diag}}) + L_{\text{off}} E, \quad (22)$$

then solve $(I - M)u = R f_{\text{out}}$.

The extension E is split in two. E_{self} holds the non-routed rows; the routed rows of the diagonal block are zeroed. E_{cross} holds the routed rows, which interpolate the target branch at the closest point of the rotated node.

The right-hand side is the source branch's f at the outer rows. On a routed row it is overwritten by the *target* branch's f at the target closest point, for the same reason as above.

The solve is a sparse direct solve while the unknown count stays at or below 25 000, and matrix-free restarted GMRES otherwise, with the $(I - L_{\text{diag}})$ diagonal preconditioner, tolerance 10^{-10} , restart 80 and 200 outer cycles. A GMRES failure is an error, not a warning. The residual of the returned solution is recomputed and stored.

5.3 What the scripts refuse to do

Both tests are written so that a silently thinned band is impossible. Each of the following raises an error rather than a warning:

- an interpolation stencil of a closest point leaves the band;
- a cross-junction stencil leaves the *other* band;
- a cross-junction stencil reads a row whose Laplacian stencil is incomplete;
- a row sum of the extension differs from one by more than 10^{-10} ;
- the band or routing closure does not converge in the allowed sweeps;
- the linear solve does not reach its residual target;
- a d_{k2} probe is still unusable after the fallback probe.

The row sums and the linear residual are reported per level, so the tables below can be read against them.

6 How we measure the error

The reported quantity is the surface L_∞ error of the *interpolant*, not of the band values.

On the ellipsoid, each branch is sampled at the midpoints of a uniform partition of its polar-angle interval, and around each parallel with a count set by that parallel's own circumference. No sample falls exactly on the cut circle or at a pole. A sample whose interpolation stencil is not entirely inside the band is dropped and counted. The script also records *where* the worst point sits, as an arclength from the cut circle in multiples of h . That is the test of whether the crease is what hurts the solve.

On the tube, each strip is sampled on a uniform (θ, s) grid with about two samples per h in each direction, and *both* edge curves are included. The script also reports the trace discrepancy of the two branches on each corner curve.

The ellipsoid test reports four further cross-branch measures at the nodes the two bands have in common:

measure	definition
gap	$\max e_A - e_B $ at the shared rotated nodes
vtx	$\max u_A - u_B $ on the cut circle
branch	$\max(e_A , e_B)$ at the same nodes
allgap	$\max e_A - e_B $ over the whole overlap

Each branch is compared against the value its *own* unknown stands for. At a crease the raw difference of the two unknowns is $O(h)$ whatever the scheme does, because the two unknowns stand for u at different points. What has to be compared is each branch's own error.

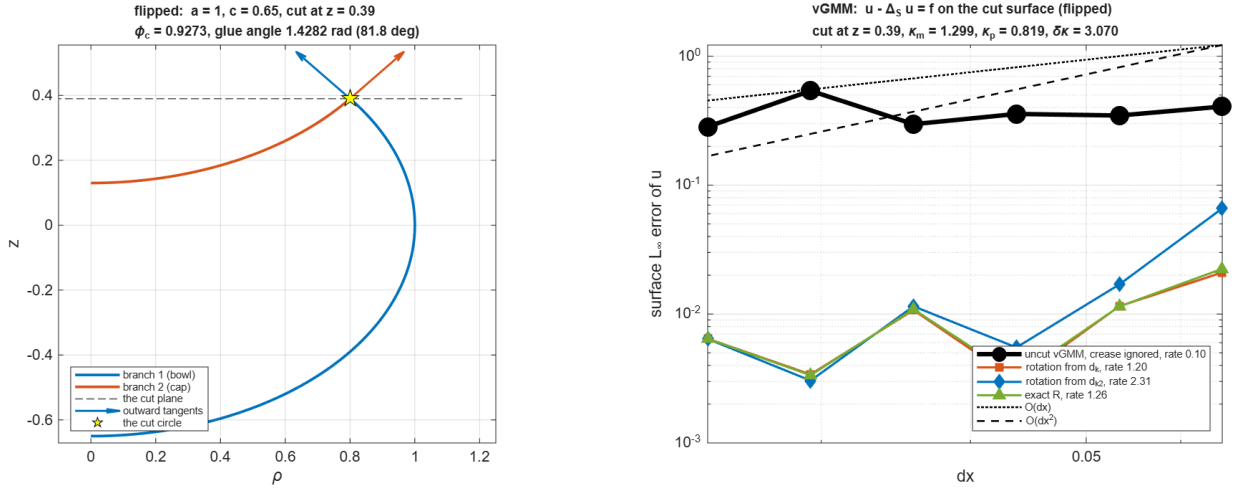


Figure 2: The cut at $z = 0.39$, the one with the largest curvature mismatch of the four. Left: the meridian half plane. Branch 1 (blue) is the bowl. Branch 2 (red) is the cap, reflected down across the cut plane, so it runs *inside* the bowl and the two meet at the star in a crease of 81.8° . The two arrows are the outward tangents the glue rotation is built from. Right: the surface L_∞ error against h . The heavy black line is the uncut solve with the crease ignored; it is flat. The three glued variants lie on top of one another, and their sawtooth is the grid-alignment scatter, not a failure of the solve.

7 Results: the flipped cap on an ellipsoid

7.1 The geometry of one cut

7.2 The glue is what makes the solve converge

The clearest result of this test is the gap between the baseline and the glued solves. Table 2 gives the cut at $z = 0.39$ in full.

Table 2: Surface L_∞ error of u , flipped ellipsoid, cut at $z = 0.39$ ($\varphi_c = 0.9273$, $\kappa_m = 1.2986$, $\kappa_p = 0.8187$, glue angle 1.4282 rad = 81.8° , $\delta\kappa = 3.0703$). The rate under each error is the level-to-level ratio. Only the fitted slope is a rate; see Section 7.3.

h	uncut, crease ignored		rotation from d_k		rotation from d_{k2}		exact R	
	error	rate	error	rate	error	rate	error	rate
0.0650	4.071e-01	—	2.101e-02	—	6.599e-02	—	2.244e-02	—
0.0533	3.467e-01	0.81	1.149e-02	3.04	1.697e-02	6.84	1.145e-02	3.39
0.0437	3.566e-01	-0.14	3.057e-03	6.67	5.507e-03	5.67	3.261e-03	6.33
0.0358	2.970e-01	0.92	1.074e-02	-6.33	1.149e-02	-3.70	1.089e-02	-6.07
0.0294	5.391e-01	-3.00	3.403e-03	5.79	3.046e-03	6.69	3.356e-03	5.93
0.0241	2.836e-01	3.24	6.445e-03	-3.22	6.414e-03	-3.75	6.436e-03	-3.28
fitted	0.10		1.20		2.31		1.26	

Three things are worth reading off it.

The baseline does not converge. The uncut vGMM solve, which applies the closest point method to the creased surface as if the crease were not there, has a fitted slope of 0.10. Its error stays between 0.28 and 0.54 at every grid spacing. It is $O(1)$. That is the expected behaviour: at the crease the closest point map of the union surface is not single valued, so the extension has no consistent footpoint, and refining the grid does not repair it.

The glue converges, at about first order. The glued solves drop the error by about 1.7 orders of magnitude, to 6.4×10^{-3} at the finest level, and the fitted slope of the exact rotation is 1.26. This is well short of the second-order rate a smooth closest point solve would give, and it is what Equation (29) predicts: with $\delta\kappa$ (and hence D) a definite nonzero number, the term hD sits at first order and dominates h^2 .

The three rotations are the same solve. At the three finest levels the three variants agree to within 7 per cent, and at the finest level to within 0.5 per cent: 6.445, 6.414 and 6.436×10^{-3} . The differences between their fitted slopes (1.20, 2.31, 1.26) are therefore *not* a difference between the schemes. They are an artifact of where the scatter falls on the coarse levels, where d_{k2} starts three times worse than the other two. The glue rotation does not have to be exact for this solve: it has to be good enough, and both estimators are.

7.3 Why a level-to-level rate is meaningless here

The per-level rates in Table 2 run from -6.33 to $+6.84$. The error is not monotone: it falls to 3.3×10^{-3} at $h = 0.0437$, rises to 1.1×10^{-2} at $h = 0.0358$, falls again, and rises again at the finest level.

This is the behaviour the script's own header predicts, and it is not noise in the solver. The residual of the solve is below 10^{-10} at every level and the extension row sums are correct to 10^{-15} . The size of the surface error is set by the net *signed* junction residual summed around the cut circle. That sum is a near cancellation between the two branches: each contributes a systematic total of one sign and the other the opposite sign. What survives depends on where the cut circle happens to fall between grid planes, which changes with h in a way that has nothing to do with accuracy.

The position of the worst point confirms this. The script reports its distance from the cut circle in multiples of h :

	h	0.0650	0.0533	0.0437	0.0358	0.0294
worst point, in units of h from the crease		0.9	0.3	10.5	0.5	0.6

At five of the six levels the worst point sits within one grid cell of the crease, which says the crease is what limits the solve. At $h = 0.0437$, the one level where the error dips, it has moved ten cells away: at that grid alignment the junction residual cancelled, and what was left was the ordinary interior error.

In the plane the same cancellation exists. There the junction is two points instead of a circle, the sums run over $O(1)$ rows instead of $O(1/h)$ rows, and the scatter never surfaces.

A caveat on the fit. The script's default is ten grid spacings, and the reason is precisely to average over this scatter. The runs in this report use *six*, because the finer levels of the default list cost more than the time available. The fitted slopes here are therefore less well determined than the script intends. They should be read as "first order, not second", not as three-digit numbers.

7.4 The cross-branch measures

The four cross-branch measures behave better than the surface error, because they do not depend on the signed cancellation around the circle. For the cut at $z = 0.39$ the fitted slopes are:

measure	what it is	d_k	d_{k2}	exact R
gap	$\max e_A - e_B $ at the shared rotated nodes	2.90	4.68	1.97
vtx	$\max u_A - u_B $ on the cut circle	1.96	3.26	1.76
branch	$\max(e_A , e_B)$ at the same nodes	1.35	3.49	1.21
allgap	$\max e_A - e_B $ over the whole overlap	2.90	4.68	1.98

The two branches agree with each other at close to second order (**gap** and **allgap**, 1.97 and 1.98 for the exact rotation), and they agree on the cut circle itself at 1.76. The **branch** measure, which is each branch’s own error at those nodes, tracks the surface error at 1.21. So the glue is *consistent* across the junction to better than the order at which either branch is *accurate*. The limit is not the transmission; it is the crease.

7.5 The estimated rotations

The fitted rates of the angle error $|\vartheta - \vartheta_{\text{exact}}|$ at this cut are

$$d_k: 2.13, \quad d_{k2}: 3.60.$$

Both are better than the nominal order of the underlying difference. The reason is the averaging: each closest point difference is resolved in its own node’s meridian frame and then averaged over the whole cut circle, so the leading error term, which is odd around the circle, cancels.

In absolute terms, however, d_k is the *better* estimator over the whole range tested:

h	0.0650	0.0533	0.0437	0.0358	0.0294	0.0241
d_k	1.10e-02	4.43e-03	2.98e-03	2.61e-03	1.45e-03	1.15e-03
d_{k2}	1.62e-01	8.42e-02	2.88e-02	1.75e-02	7.26e-03	5.27e-03

d_{k2} starts two orders of magnitude worse and is still 4.6 times worse at the finest level. Its steeper slope means it would overtake eventually, but it has not done so on any grid in this study. Neither difference reaches the solution: at the finest three levels both estimators give the same surface error as the exact rotation.

7.6 All four cuts, against the curvature mismatch

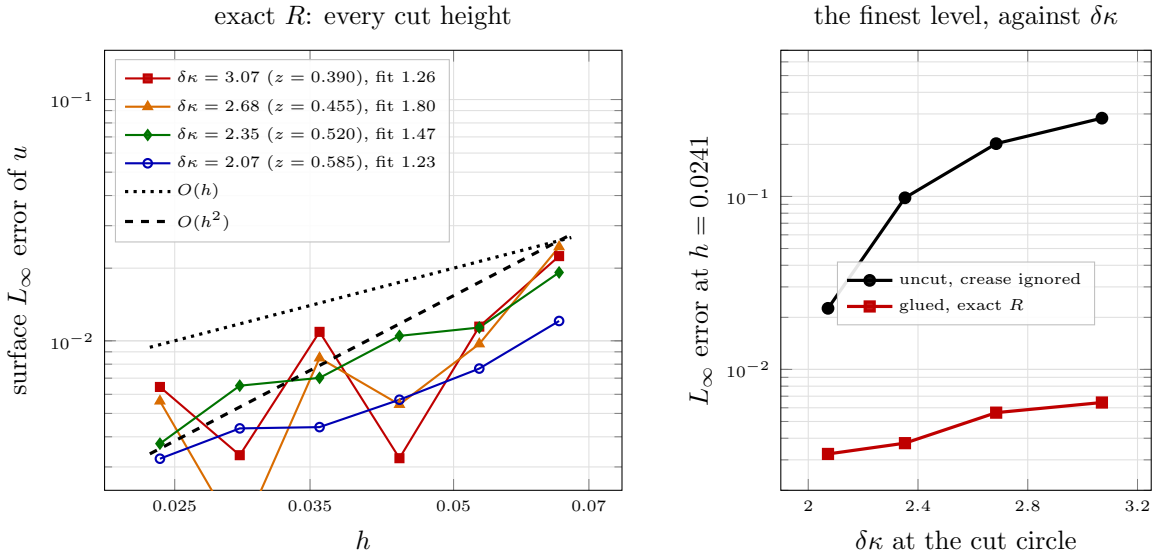


Figure 3: All four cut heights. Left: the surface L_∞ error of the glued solve with the exact rotation, against h . The uncut baseline is left out of this panel, because it is two orders of magnitude higher; it appears in Figure 2 and in the right panel here. All four glued curves sit between the $O(h)$ and $O(h^2)$ guides, and all four zigzag. Right: the error at the finest level, against the curvature mismatch $\delta\kappa$. Both the baseline and the glued solve order monotonically by $\delta\kappa$, although the *fitted slopes* do not — see the text below.

The four cut heights span crease angles from 34.9° to 81.8° , and with them a range of curvature mismatch. Table 3 collects them.

Table 3: All four cuts of the flipped ellipsoid. $\delta\kappa$ falls as the cut moves from the equator towards the pole. Each entry is the fitted slope over six levels, with the error at the finest level ($h = 0.0241$) beside it.

z	φ_c	crease	κ_m	$\delta\kappa$	uncut, crease ignored		d_k		d_{k2}		exact R	
					rate	error	rate	error	rate	error	rate	error
0.390	0.9273	81.8°	1.2986	3.0703	0.10	2.84e-01	1.20	6.45e-03	2.31	6.41e-03	1.26	6.44e-03
0.455	0.7954	67.1°	1.0970	2.6849	0.59	2.02e-01	1.78	5.60e-03	2.20	5.80e-03	1.80	5.62e-03
0.520	0.6435	52.0°	0.9220	2.3525	0.33	9.82e-02	1.47	3.74e-03	1.89	3.70e-03	1.47	3.75e-03
0.585	0.4510	34.9°	0.7738	2.0720	1.00	2.25e-02	1.22	3.26e-03	1.22	3.28e-03	1.23	3.24e-03

The error level orders by $\delta\kappa$. The fitted rate does not. This is the main qualification on this test. At the finest level the error of the glued solve falls monotonically as $\delta\kappa$ falls:

$$6.44, \quad 5.62, \quad 3.75, \quad 3.25 \quad (\times 10^{-3}) \quad \text{for} \quad \delta\kappa = 3.07, 2.68, 2.35, 2.07.$$

The uncut baseline does the same, and far more strongly: from 2.84×10^{-1} down to 2.25×10^{-2} , a factor of 12.

The fitted slopes, by contrast, are 1.26, 1.80, 1.47 and 1.23. They do not order by $\delta\kappa$, and they do not order by anything else either. Six levels are not enough to separate a trend in the rate from the grid-alignment scatter of Section 7.3. The defensible statement from this study is therefore:

Every cut converges at a fitted rate between 1.2 and 1.8. None reaches second order.
 The size of the error at a fixed grid spacing grows with $\delta\kappa$, as Equation (29) says it should — through the constant in $C h D$, not through the exponent.

That is how a bound behaves when its junction term is first order with a $\delta\kappa$ -dependent constant. Changing $\delta\kappa$ moves the curve up and down. It would change the *rate* only if $\delta\kappa$ reached zero. The unflipped cut, where $\delta\kappa$ vanishes identically, is the test of that. It is the `flipped = false` branch of the script, and it is not run here.

Where the error sits. The script reports the distance of the worst point from the cut circle, in units of h . At the two sharpest cuts that distance is within one cell at nearly every level. At the mildest cut, $z = 0.585$, it is 11.5, 14.1 and 16.6 cells at the three coarsest levels, and it moves to the crease only at the three finest. At a mild crease the junction is not the worst thing in the solve until the grid is fine enough for the interior error to fall below it.

The estimated rotations, across all cuts. At every cut the three variants agree at the finest level to within 1 per cent. The fitted rates of the angle error are 2.13, 2.30, 2.36 and 1.98 for d_k , and 3.60, 3.25, 3.76 and 2.93 for d_{k2} . Both estimators beat the nominal order of their underlying difference at every cut, and neither difference reaches the solution.

8 Results: the half-twisted rectangular tube

8.1 The bands and the routed rows

The routed rows are not a thin shell. Table 4 gives the counts. At every level about one unknown in five is a row whose value is supplied by the glue rather than by its own branch. That is the reason the junction term in Equation (23) is not a local perturbation on this surface: there are two closed junction curves, and every one of their neighbourhoods is routed.

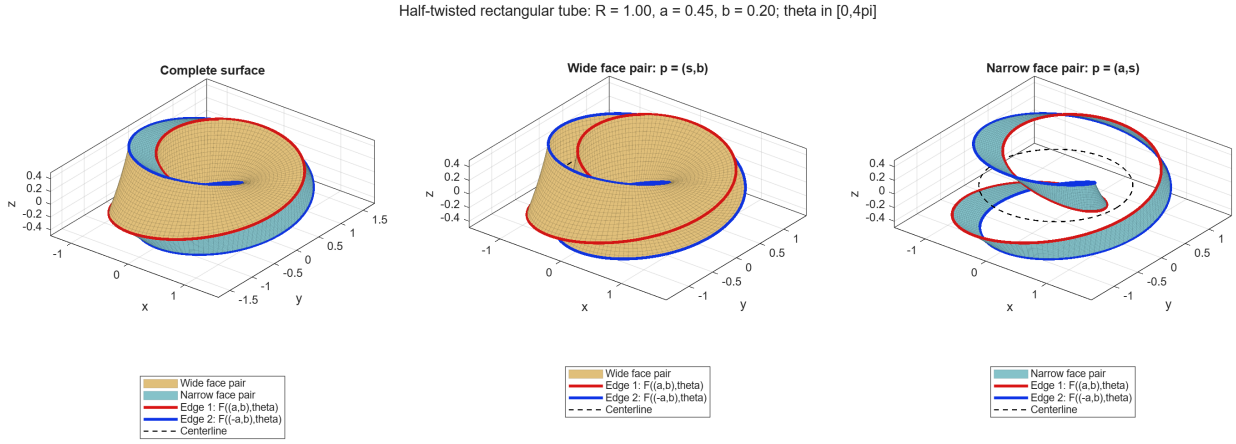


Figure 4: The half-twisted rectangular tube and its two face-pair strips. Left: the complete surface, which is a closed orientable torus. Middle: the wide strip, $p = (s, b)$ with $s \in [-a, a]$. Right: the narrow strip, $p = (a, s)$ with $s \in [-b, b]$. Each strip covers its face pair once over $\theta \in [0, 4\pi]$, and each is an annulus, not a Möbius strip. The red and the blue curve are the two closed corner curves where the strips meet. The cross-section sides are straight, but the faces they sweep are not planar.

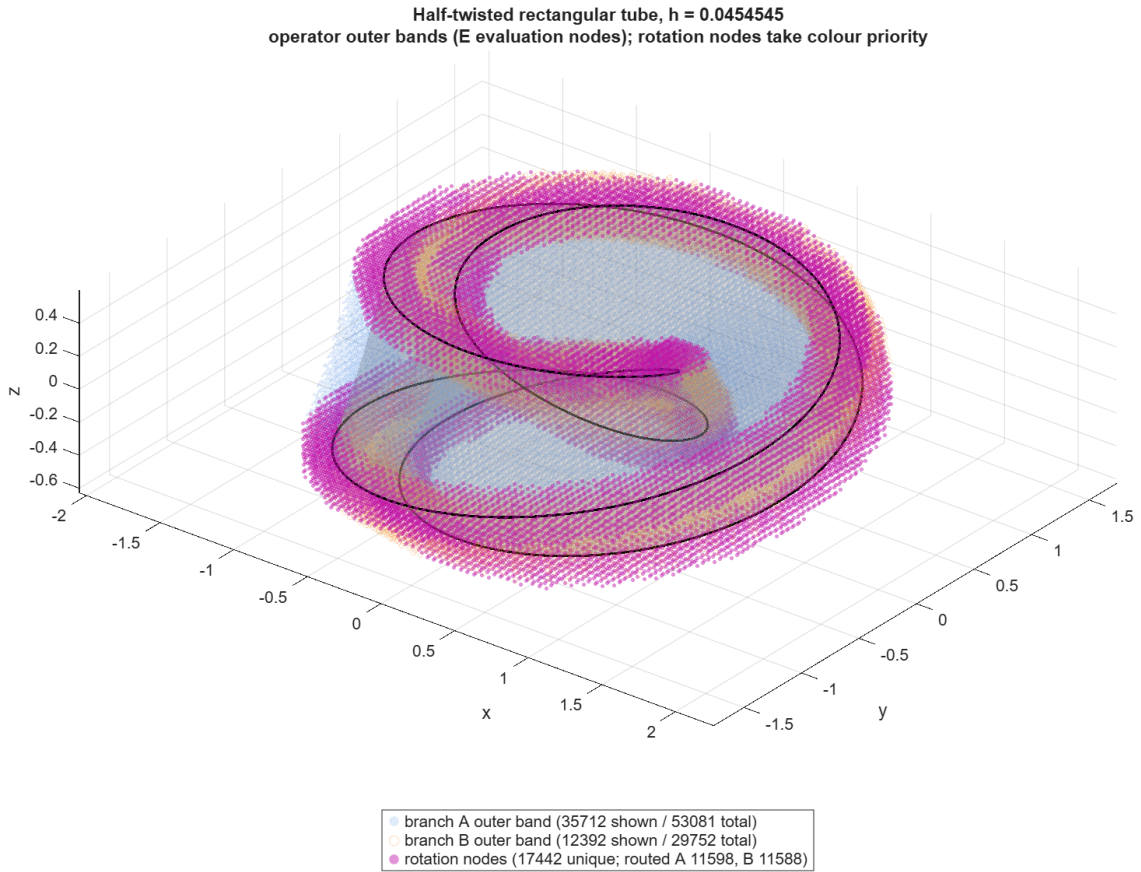


Figure 5: The two narrow bands at $h = 1/22$, and the nodes the glue rotates. Blue dots are the outer band of branch A, orange rings the outer band of branch B, and magenta the rows that are routed across a corner curve. A node that lies in both bands carries both markers, because it carries two independent unknowns.

Table 4: Problem size and solver behaviour, half-twisted rectangular tube. Routed rows are summed over both branches and both corner curves. The row-sum error is $\max_i |\sum_j E_{ij} - 1|$.

h	unknowns A	unknowns B	routed rows	GMRES iter	rel. residual	row-sum error
0.05000	32 393	17 391	21 029	722	9.7e-11	1.3e-15
0.04000	49 040	25 556	26 686	840	9.9e-11	1.6e-15
0.03333	68 599	34 963	31 930	1203	9.9e-11	1.6e-15
0.02778	96 784	48 138	37 808	1544	9.9e-11	1.6e-15
0.02273	141 729	69 344	46 307	2276	1.0e-10	1.6e-15

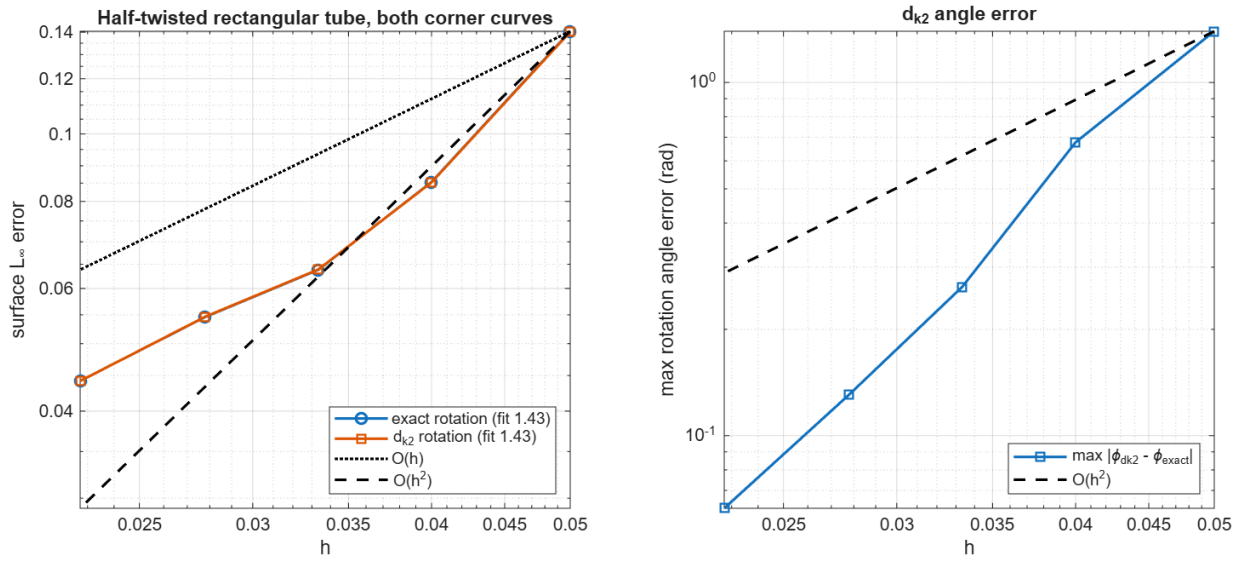


Figure 6: Left: the surface L_∞ error of both rotations, with $O(h)$ and $O(h^2)$ guides. The two curves lie on top of one another. The error starts between the guides and bends towards $O(h)$ as h falls. Right: the worst d_{k2} rotation angle error, against an $O(h^2)$ guide. It falls faster than second order.

Table 5: Surface L_∞ error of u , combined over both strips, half-twisted rectangular tube with $R = 1$, $a = 0.45$, $b = 0.20$. Rates use the actual h ratios.

h	exact rotation		d_{k2} rotation		d_{k2} worst angle error (rad)
	error	rate	error	rate	
1/20	1.4041e-01	—	1.4018e-01	—	1.3967e+00
1/25	8.5293e-02	2.23	8.5201e-02	2.23	6.7693e-01
1/30	6.3814e-02	1.59	6.3876e-02	1.58	2.6388e-01
1/36	5.4570e-02	0.86	5.4621e-02	0.86	1.3056e-01
1/44	4.4203e-02	1.05	4.4186e-02	1.06	6.2282e-02
fitted	1.429		1.426		4.04

8.2 Convergence of the L_∞ error

The fitted rate is 1.43, and it is not an asymptotic order. Table 5 gives a least-squares slope of 1.429 for the exact rotation. That number alone is misleading, because the level-to-level rates fall steadily:

$$2.23, \quad 1.59, \quad 0.86, \quad 1.05.$$

The solve starts close to second order on the coarsest pair and settles near first order on the finest pairs. Figure 6 shows the same thing: the error curve starts below the $O(h^2)$ guide and bends towards the $O(h)$ guide. The fitted 1.429 is an average over a range in which the rate is still changing, not the order the scheme will hold at smaller h . The honest statement is that the observed behaviour is consistent with a first-order junction term taking over from a second-order interior term, which is what Equation (23) predicts when D and J_2 do not vanish.

Unlike the ellipsoid, this surface has no grid-alignment scatter: the error is monotone in h at every level. The two junction curves are not plane circles, and the glue angle varies along them, so the signed junction residual does not add up to a near cancellation that depends on grid position.

Cross-check against the recorded run. The notes in `angle.rotation.glue.md` record a run at $h \in \{1/25, 1/30, 1/40\}$. The three levels this study shares with it reproduce it to every printed digit: $8.5293\text{e-}2$ and $6.3814\text{e-}2$ for the exact rotation, $8.5201\text{e-}2$ and $6.3876\text{e-}2$ for d_{k2} , and the level rate 1.59 for the pair $1/25 \rightarrow 1/30$.

8.3 The estimated rotation costs nothing here

The two rotations are indistinguishable in the solution. The largest relative difference over all five levels is 0.16 per cent, at the coarsest level, and at the finest level the d_{k2} solve is marginally the *better* of the two (4.4186 against 4.4203×10^{-2}). That is not a real advantage; it is the size of the difference.

This holds although the d_{k2} rotation is a poor estimate on the coarse grids. Its worst angle error is 1.40 rad at $h = 1/20$, which is 80° on the worst row. The error falls at a fitted rate of 4.04, with level rates 3.25, 5.17, 3.86 and 3.69, reaching 6.2×10^{-2} rad at the finest level.

How many rows need the fallback probe. The d_{k2} source estimate is re-sampled from the deterministic outward probe $c + h\eta_s$ whenever the reflected ray is unusable. The counts are substantial and they grow with refinement:

h	probed rows, branch A	probed rows, branch B	sign reversals
1/20	2972	1789	7
1/25	3100	2248	0
1/30	3000	2609	0
1/36	3360	3107	0
1/44	3973	3854	0

So roughly one routed row in six falls back to the probe. The probe is placed with the *analytic* conormal, which is why this variant is not a geometry-free estimator, as the script states. The seven sign reversals at the coarsest level are rows where the long reflected sample misjudged which side of the edge it was on.

8.4 The traces agree at nearly second order

The two branches agree with each other on the corner curves far better than either agrees with the exact solution:

h	$\max u_A - u_B $, upper curve	$\max u_A - u_B $, lower curve
1/20	2.879e-02	2.765e-02
1/25	1.610e-02	2.123e-02
1/30	1.219e-02	1.288e-02
1/36	7.763e-03	1.144e-02
1/44	6.412e-03	7.186e-03
fitted	1.93	1.71

The trace discrepancy converges at 1.93 and 1.71, against 1.43 for the surface error itself, and this is the same pattern the ellipsoid shows. The glue transmits consistently across the junction at close to second order. What limits the solve is not the transmission.

9 What the two tests say together

9.1 The bound the two tests are read against

The surface analysis [4] gives, for the glued solve,

$$\|Q_h u_h - u\|_\infty \leq C \left\{ h^q + h^{p-1} + h(D + J_2) + h^2 + \varepsilon_R + h \rho_{f,h} \right\}, \quad (23)$$

conditional on a discrete stability assumption. With $q = 2$ and $p = 3$, and with a source-consistent forcing convention, this reads

$$\|Q_h u_h - u\|_\infty \leq C \{ h^2 + h(D + J_2) + \varepsilon_R \}. \quad (24)$$

Three constants in (24) decide what the tables can show.

D , the extrinsic junction defect. D is the supremum over the junction curve of $|b_2^\perp - b_1^\perp|$, the mismatch of only the *mixed* and *transverse* parts (b_{sr}, b_{rr}) of the second fundamental form in the matched edge frame. On the flipped ellipsoid the meridian and the parallel are the principal directions of both branches, so $b_{sr} = 0$ on both and

$$D = 2\kappa_m. \quad (25)$$

The along-edge curvature κ_p drops out of D , because the transmission conditions already match U_{ss} .

$\delta\kappa$, the quantity the tables report. The ellipsoid script reports instead

$$\delta\kappa = \|\Pi_1 - R \Pi_2 R^\top\|_F, \quad (26)$$

the full matched-frame mismatch, which [4] calls a valid but less sharp replacement for D . On this family $\delta\kappa$ runs 18 to 34 per cent above D , and it is monotone in the same direction, so ranking the cuts by either gives the same order.

The script builds $\delta\kappa$ rather than assuming it. Both branches are pieces of the same ellipsoid, met at the same φ_c , so they share

$$\kappa_m = \frac{ac}{G^{3/2}}, \quad \kappa_p = \frac{c}{a\sqrt{G}}, \quad G(\varphi) = a^2 \cos^2 \varphi + c^2 \sin^2 \varphi, \quad (27)$$

and the mismatch collapses to $|n_1 - Rn_2| \sqrt{\kappa_m^2 + \kappa_p^2}$ with n the inward unit normals. On the *unflipped* cut, R is the identity and $n_2 = n_1$, so $\delta\kappa$ vanishes identically and the glued solve should recover the second-order rate of the uncut one. On the *flipped* cut, R carries n_2 to $-n_1$, so $|n_1 - Rn_2| = 2$ and

$$\delta\kappa = 2\sqrt{\kappa_m^2 + \kappa_p^2} \quad (\text{flipped}) \quad (28)$$

at every cut height. It varies only through where on the meridian the cut falls.

J_2 , the intrinsic junction defect. J_2 is $[U_{rr}] = (k_2 - k_1)U_r - [f]$, with k_i the geodesic curvature of the junction curve inside branch i . On the flipped ellipsoid it vanishes, and *both* of its terms vanish separately. The two branches are mirror images across the cut plane, so $k_2 = k_1$. And u and f are the same functions of (φ, ϑ) on both branches, so $[f] = 0$.

That is a different route to $J_2 = 0$ than the two-sphere example of [4], where the two terms are individually nonzero and cancel. On the flipped ellipsoid the whole junction defect is therefore *extrinsic*, and the bound reads

$$\|Q_h u_h - u\|_\infty \leq C\{h^2 + hD + \varepsilon_R\}. \quad (29)$$

ε_R , the rotation error. This is the term the estimated rotations pay and the exact rotation does not. It is the only term that separates the three variants of the ellipsoid test, and the only term that separates **exact** from **dk2** on the tube.

$\rho_{f,h}$, the forcing residual. This is why both scripts take f at the *rotated target* on a routed row. Evaluating the forcing on the opposite side without correcting a trace jump can leave $\rho_{f,h} = O(1)$ instead of $O(h)$. That would promote $h\rho_{f,h}$ from $O(h^2)$ to $O(h)$ and put it on a level with the junction term itself. It is safe in both tests only because $[f] = 0$ by construction.

9.2 The two surfaces are different tests of the same thing

The two surfaces differ in nearly every way that could matter, which is what makes the agreement between them useful.

	flipped ellipsoid	half-twisted tube
junction	one plane circle	two closed non-planar curves
glue angle	one angle for the whole circle	varies along each curve
operator	vGMM, $M = EL - \gamma(I - E)$	diagonally split ICPM
solve	matrix-free GMRES	direct, then matrix-free GMRES
J_2	zero, both terms separately	nonzero in general
estimators	d_k and d_{k2} , fully geometry-free	d_{k2} , with analytic axis and probes
scatter in h	large	none
fitted rate	1.23 to 1.80	1.43

They agree on the three things that matter. First, the rate is between first and second order and nearer to first. Second, the glue rotation may be estimated rather than computed, at no cost in the solution. Third, the junction transmission converges faster than the solution does.

9.3 Why the ellipsoid scatters and the tube does not

This is the sharpest contrast in the study, and it is worth stating plainly, because it changes how each table must be read.

On the ellipsoid the junction is a circle in a plane, and the glue rotation is one angle for every point of it. Each branch therefore contributes a junction residual of one sign at every azimuth, and the two contributions very nearly cancel. What survives is set by where the circle falls between grid planes. That changes with h in a way that has nothing to do with accuracy, so neighbouring levels differ by a factor of several.

On the tube neither corner curve is planar, and the twist makes the glue angle vary along both. The residuals do not line up in azimuth, nothing near-cancels, and the error is monotone in h at every level.

The practical consequence: a convergence study on a surface of revolution cut by a plane needs many more levels than one on a generic surface. The ellipsoid script's choice of ten grid spacings is not conservatism.

9.4 What would settle the open question

The one experiment that would separate the two candidate explanations is already in the script and was not run here: the *unflipped* cut.

On the unflipped surface the two pieces join smoothly, the exact glue rotation is the identity, and $\delta\kappa$ vanishes identically. The bound then reads $C\{h^2 + \varepsilon_R\}$, and the glued solve should recover the second-order rate of the uncut one. If it does, then the order lost on the flipped surface is the *crease*, and the glue itself is second-order accurate. If it does not, then the glue construction costs an order by itself, independently of the geometry it is applied to.

Nothing in this report distinguishes those two possibilities, because every cut studied here has $\delta\kappa$ between 2.07 and 3.07 and none has $\delta\kappa = 0$.

Two smaller follow-ups are worth the same note:

- **More levels on the ellipsoid.** The script's own default of ten would make the four fitted rates comparable with one another. At six levels they are not.
- **Finer levels on the tube.** Its level rates were still falling at $h = 1/44$. One or two more levels would show whether they settle at first order or continue down.

9.5 A note on the construction, separate from the results

The (cos, sin) construction of the rotation is a correctness and reproducibility choice, not an accuracy one. The measured difference against the `atan2` path is a few units in the last place, and the planar companion experiment shows it does not reach the computed solution. What it does buy is exact: the two branches' rotations at a junction come out as bit-for-bit transposes, there is no branch cut at $\pm\pi$ for a stored angle to fall across, and every operation is one that IEEE 754 pins down. Those properties hold regardless of the rates in this report.

10 How to reproduce

All of the numbers in this report come from the two scripts named below. Run them from their own directory. No argument is needed for the first one.

```
cd cp_matrices/examples_2026/3D_surface
```

```
matlab -batch "example_ellipsoid_flipped_cap_branch_consistency_vgmm"
```

```
matlab -batch "example_half_twisted_rectangular_tube_rotation_convergence(1./[20 25 30 36 44])"
```

10.1 Two changes were needed to produce these runs

A bug stops the ellipsoid script after its first cut. In `example_ellipsoid_flipped_cap_branch_consistency_vgmm`, the local function `error_surface_figure` is declared with twenty parameters, ending `geoname`, `dkappa`. Its body then uses `thetaex(1)`:

```
dkappa, thetaex(1)*180/pi, dx)}, 'FontSize', 9);
```

`thetaex` is never passed to it. A local function in a script file does not share the script's workspace, so this raises `Unrecognized function or variable 'thetaex'`. The default `z_3d = 0.390` is not empty, so the figure is always requested, and the script always aborts after the first cut height. It therefore never reaches cuts two, three and four, nor the summary tables, nor the summary figure.

The fix is to pass the variable through. Add `thetaex` to the end of the function's parameter list, and to the end of its call. The call site already has `thetaex` in scope: it is computed once per cut height, just above.

The grid spacings were reduced. The default list is $h_k = 0.065 \times 0.82^k$ for $k = 0, \dots, 9$. At four cut heights and four solves per level, with the band size growing like h^{-2} , that is a run of roughly a day. The runs here use $k = 0, \dots, 5$, which is the same list truncated. Nothing else in the script was changed. The consequences for the fitted rates are discussed in Section 7.3.

Cut one was run separately. Because the abort above was found after cut one had completed, the first cut height comes from one process and the other three from another. The two processes used identical settings apart from `s0_list` and `z_3d`. The script's own end-of-run summary figure therefore shows only three cuts; Figure 3 in this report is drawn from the logged numbers of both processes and shows all four.

Supporting files. The two scripts share the rotation helpers and the geometry routines:

file	what it does
<code>rotations/rot2d_from_to.m</code>	the plane rotation carrying one direction onto another, by Rodas
<code>rotations/glue_rot2d.m</code>	the same, with the junction convention: outward onto inward
<code>rotations/rotate_about2d.m</code>	applies a (cos, sin) pair about a junction point
<code>rotations/rot2d_err.m</code>	the angle and matrix error of a rotation, through the relative rotation
<code>3D_surface/angle3D.m</code>	the same construction in three dimensions
<code>surfaces/cpEllipseArc.m</code>	closest point of an arc of an ellipse, with an edge flag
<code>surfaces/cpHalfTwistedRectTube.m</code>	closest point of one strip of the tube, with an edge flag
<code>surfaces/halfTwistedRectTubeFrame.m</code>	the tube parametrization and its analytic derivatives
<code>3D_surface/halfTwistedRectTubeMMS.m</code>	the manufactured solution of the tube
<code>3D_surface/halfTwistedRectTubeBands.m</code>	the grid, the two bands, and the routing targets, per level

Figures. Each script writes its own PNG files into `examples_2026/figs/`. The tube script also writes a MAT file with the full `results` structure. The two plot-only scripts `example_half_twisted_rectangular_tube` and `example_half_twisted_rectangular_tube_narrow_band.m` produce Figures 4 and 5. They solve nothing.

Theory notes. `examples_2026/angle_rotation_glue.md` carries the derivation. Section 3 covers the glue rotation step by step, and Section 10 covers the tube.

References

- [1] I. von Glehn, T. März and C. B. Macdonald. An embedded method-of-lines approach to solving partial differential equations on surfaces. 2013.
- [2] S. J. Ruuth and B. Merriman. A simple embedding method for solving partial differential equations on surfaces. *J. Comput. Phys.*, 2008. (the bandwidth formula)
- [3] T. März, C. B. Macdonald and Y. Chen. Consistency and stability of closest point iterations. Draft.
- [4] T. März, C. B. Macdonald and Y. Chen. Closest point rotation across a shared surface edge: local proofs and conditional convergence in $\mathbb{R}^3$. Draft, `output/pdf/cpm_surface_edge_analysis.pdf`. Theorem 6.4 is the bound quoted in this report; Assumption 6.3 is the discrete stability it is conditional on; Proposition 7.1 is what the uncut baseline measures.